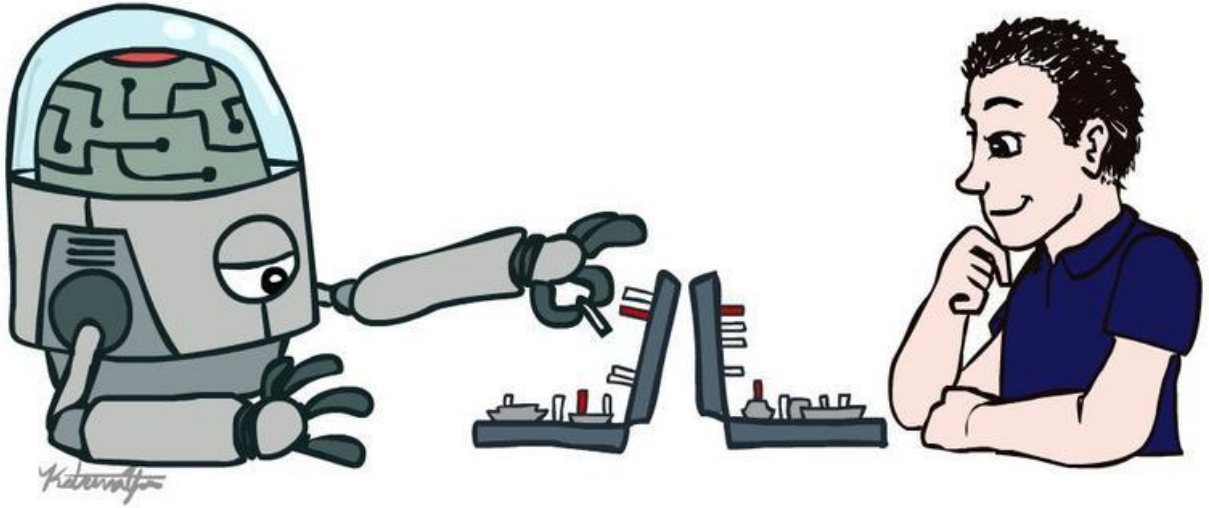
SE 210 - Special Topics in CSCI

(Artificial Intelligence)



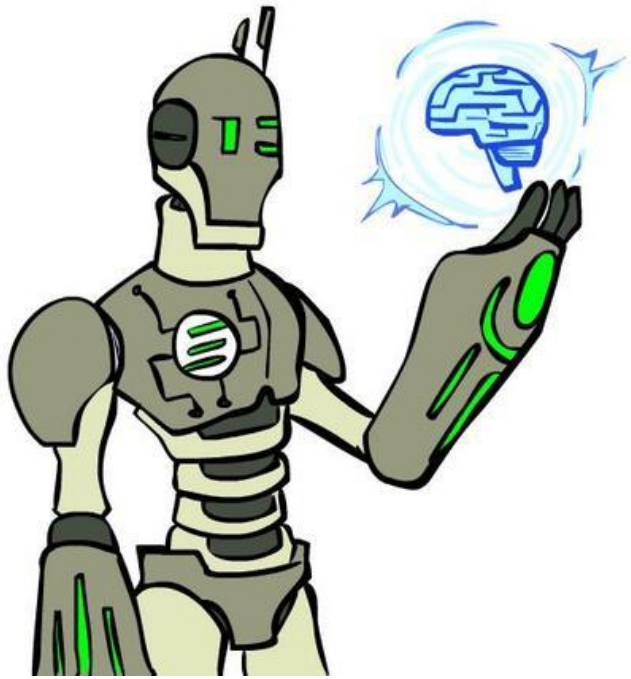
Spring 2026

University of New York, Tirana

[Credits for the slides to UC Berkeley]

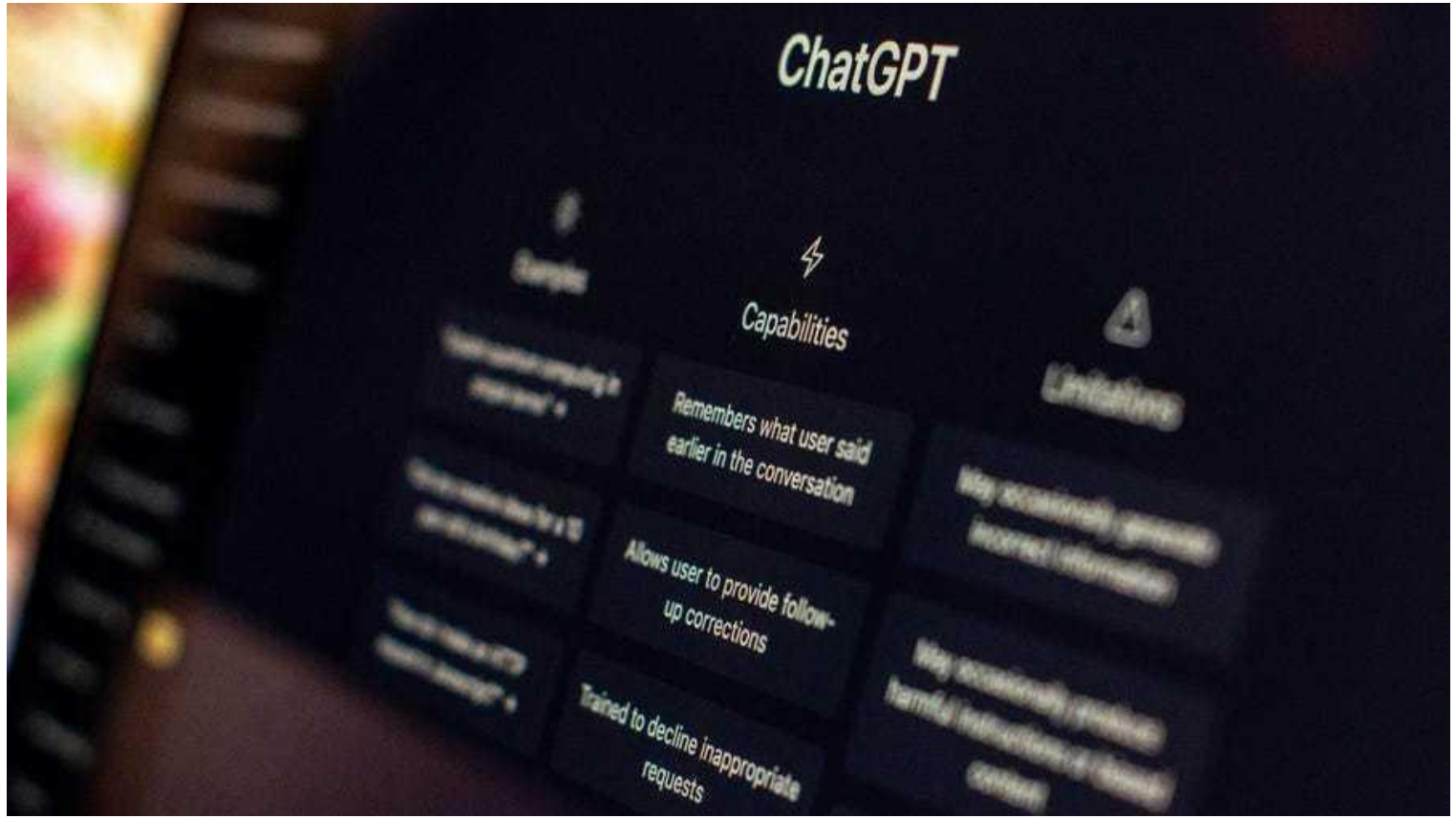
What is AI?

- What is AI?
- What can it do?
- What is this course?



AI is having real-world impact

- Public imagination
  - Textassistants

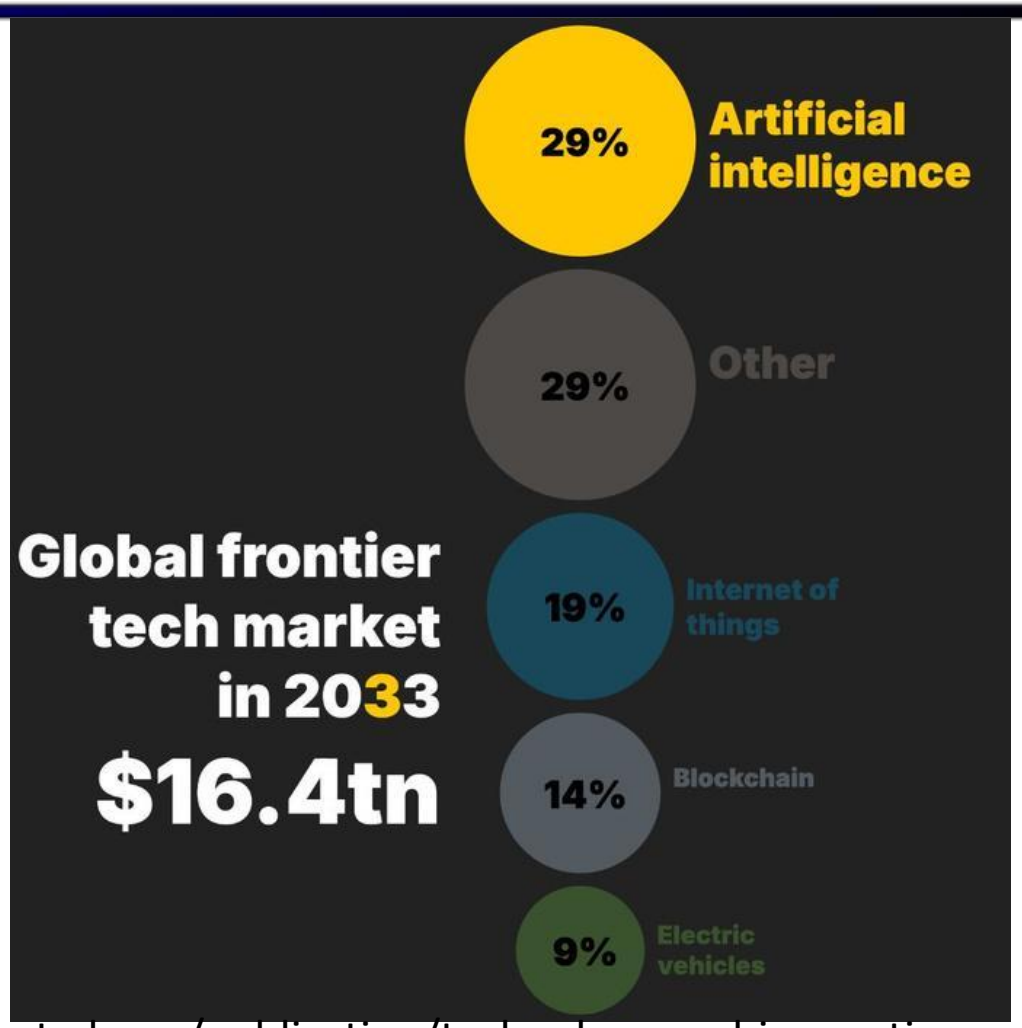


AI is having real-world impact

- Public imagination
  - Text assistants
  - Image generation

AI is having real-world impact

- Public imagination
- Economy
  - $4.8 trillion by 2033
    (25x increase in a decade)



https://unctad.org /publication/technology-and-innovation-report-2025

AI is having real-world impact

- Public imagination
- Economy Politics
-

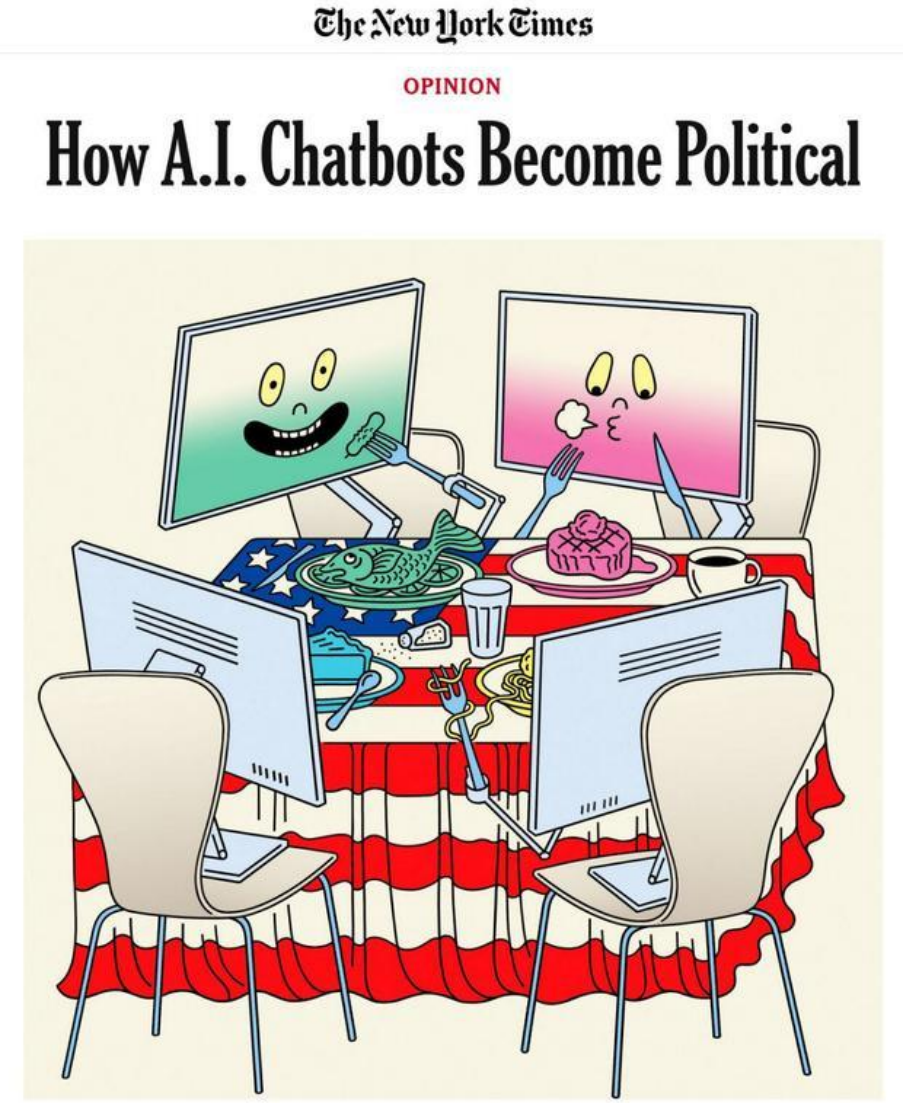
AI is having real-world impact

- Public imagination
- Economy Politics
-

https://www.nytimes.com/2025/02/11/world/europe/vance-speech-paris-ai-summit.html

AI is having real-world impact

- Public imagination
- Economy Politics
-



AI is having real-world impact

- Public imagination
- Economy Politics
- Law
-

Aug. 18, 2023, 12:18 PM; Updated: Aug. 18, 2023, 12:48 PM

AI-Generated Art Lacks Copyright Protection, D.C. Court Says (1)

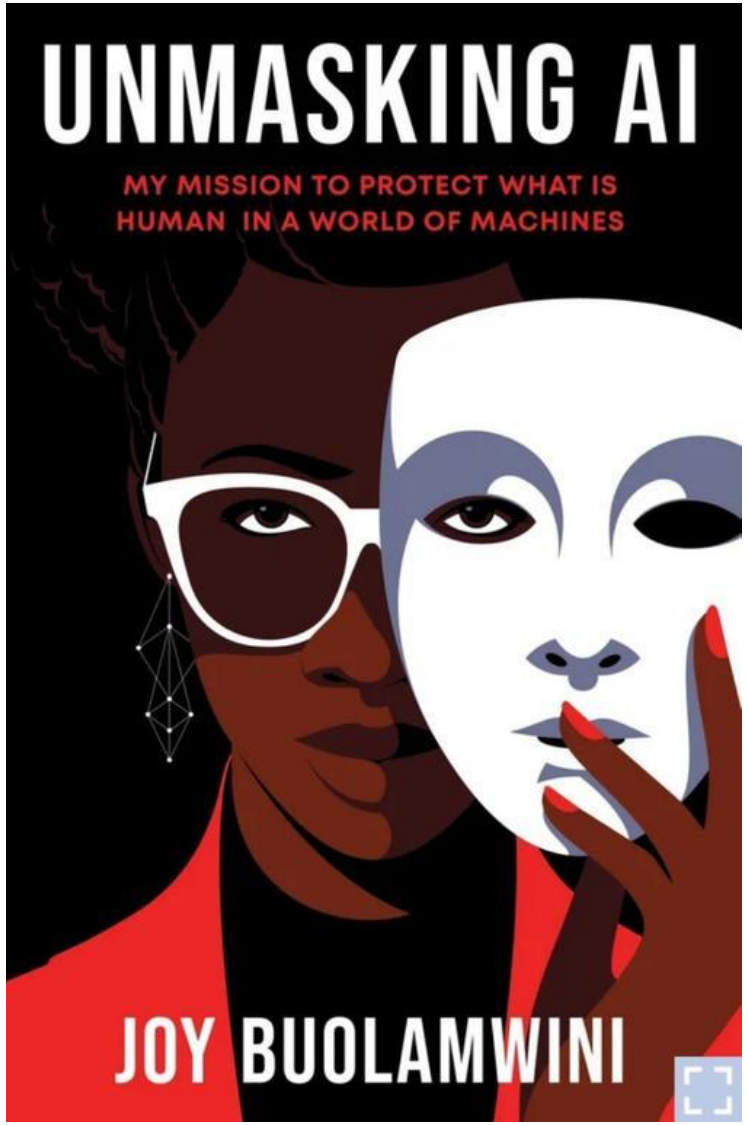
Riddhi Setty
Reporter

Isaiah Poritz
Legal Reporter

Bloomberg Law, 2023

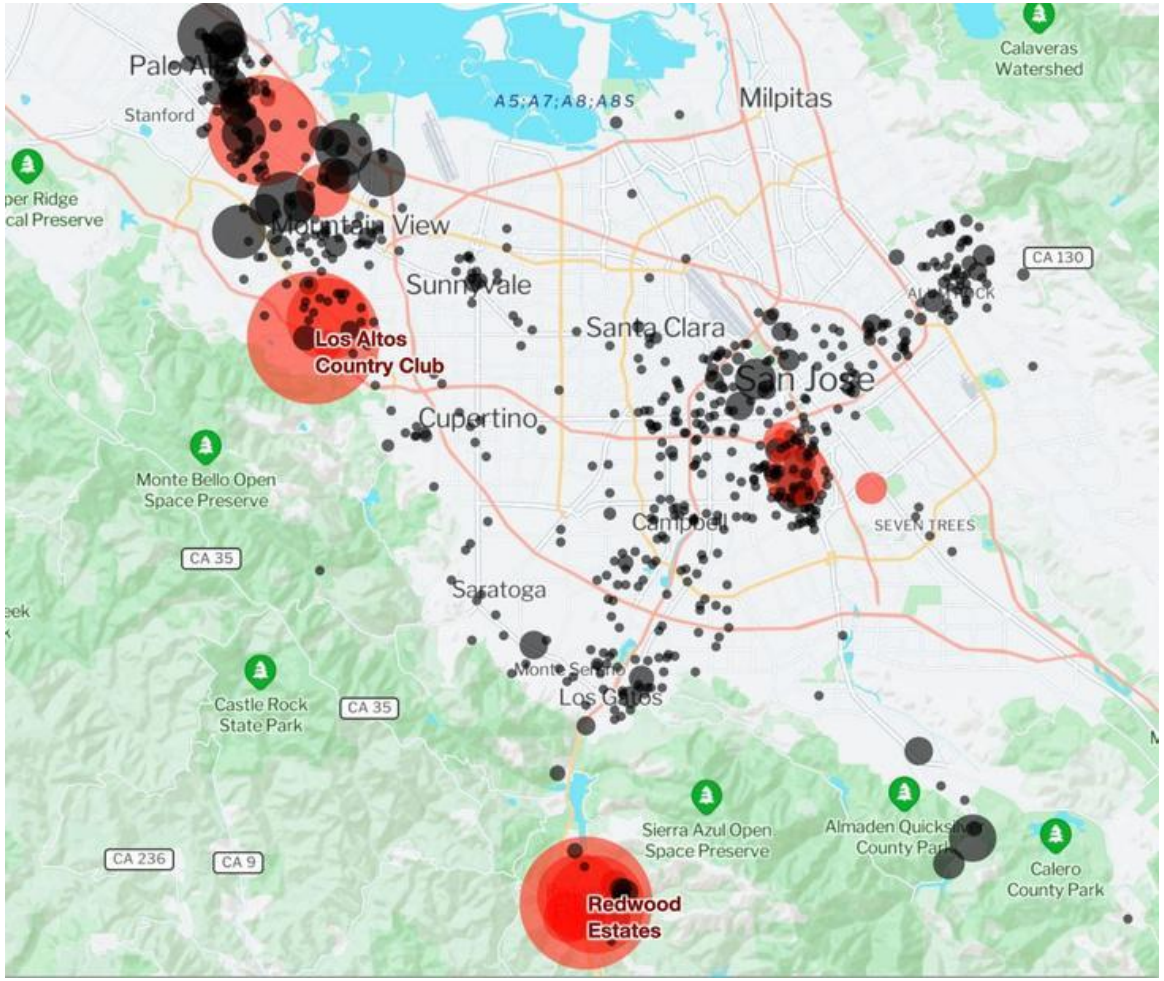
AI is having real-world impact

- Public imagination
- Economy Politics
- Law
-



AI is having real-world impact

- Public imagination
- Economy Politics
- Law
-



AI for Scaling Legal Reform: Mapping and Redacting Racial Covenants in Santa Clara County. Surani et al, 2024.

AI is having real-world impact

- Public imagination
- Economy Politics
- Law
-

https://dataplusfeminism.mit.edu

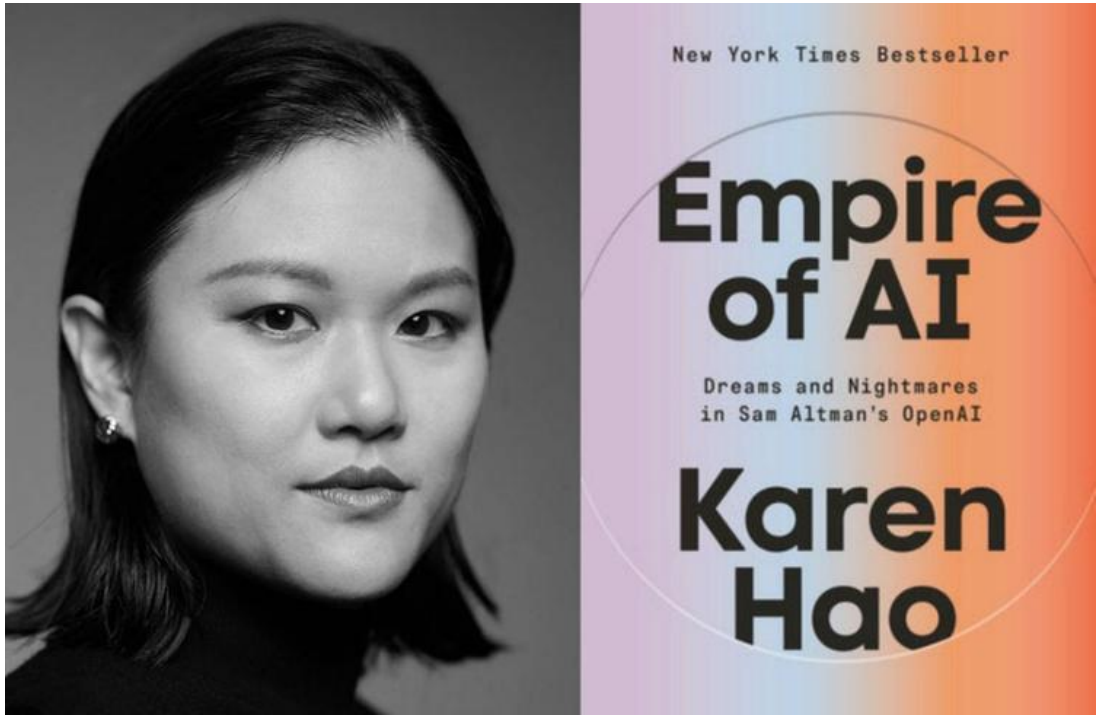
AI is having real-world impact

- Public imagination
- Economy Politics
- Law Labor
-
-

The human labor behind AI chatbots and other smart tools

Data labeling is an important step in developing artificial intelligence but also exposes the people doing the work to harmful content.

MarketWatch, 2023



AI is having real-world impact

- Public imagination
- Economy Politics
- Law Labor
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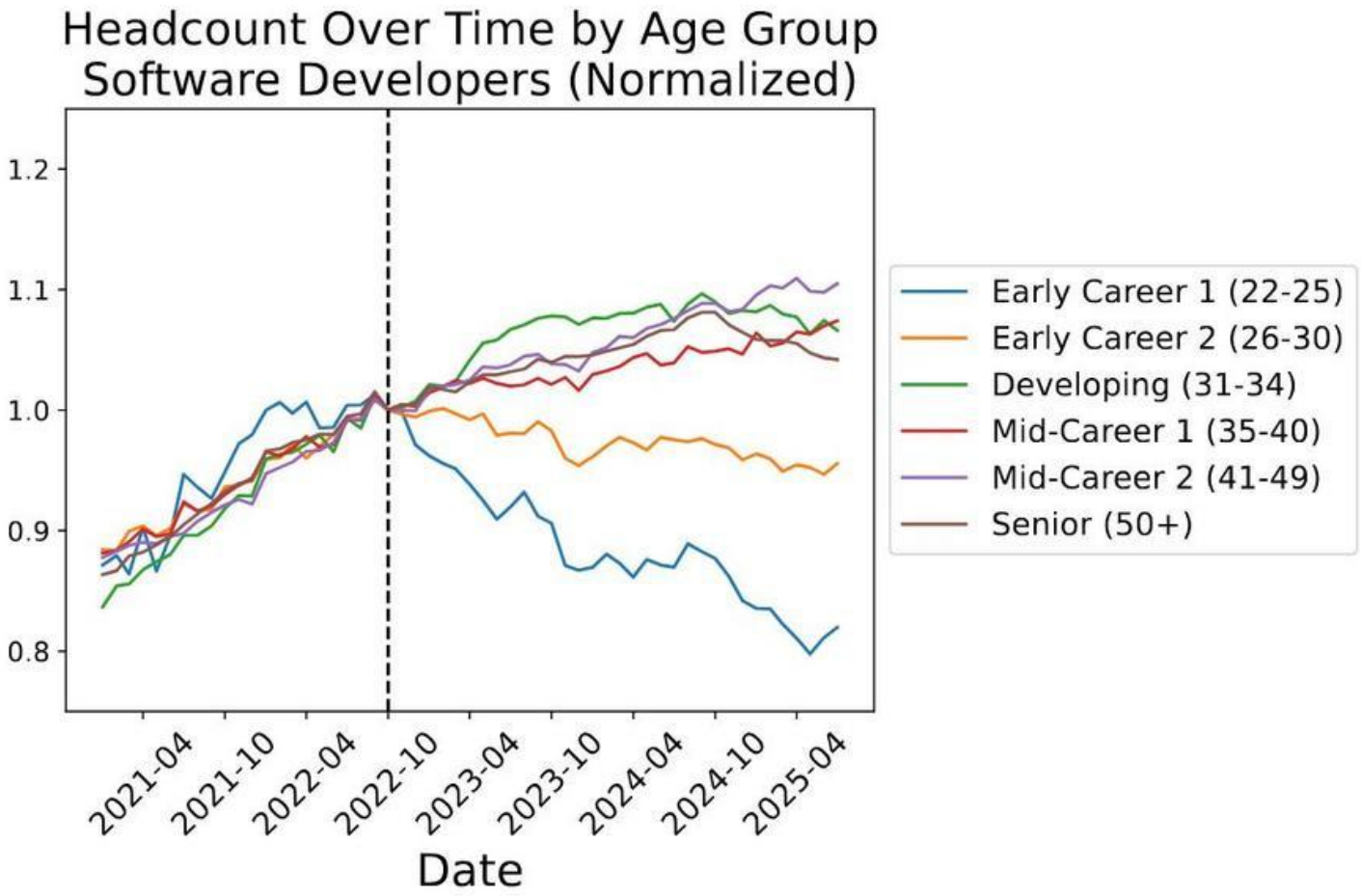
-



Stanford Digital Economy Lab, 2025

AI is having real-world impact

- Public imagination
- Economy Politics
- Law Labor Science
-
-
-
  - Chemistry and biology

Science & technology | The 2024 Nobel prizes

AI wins big at the Nobels

Awards went to the discoverers of micro-RNA, pioneers of artificial-intelligence models and those using them for protein-structure prediction

The Economist, 2024

AI is having real-world impact

- Public imagination
- Economy Politics
- Law Labor Science
-
-
-
  - Physics

AMIT KATWALA SCIENCE FEB 16, 2022 11:00 AM

DeepMind Has Trained an AI to Control Nuclear Fusion

The Google-backed firm taught a reinforcement learning algorithm to control the fiery plasma inside a tokamak nuclear fusion reactor.

PHOTOGRAPH:CURDIN WÜTHRICH, SPC/EPFL

Wired, 2022

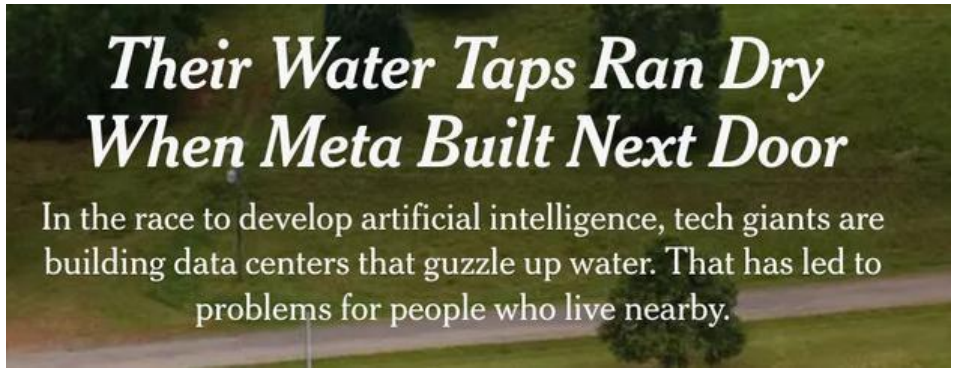
AI is having real-world impact

- Public imagination
- Economy Politics
- Law Labor Science
-
-
-
  - Conservation

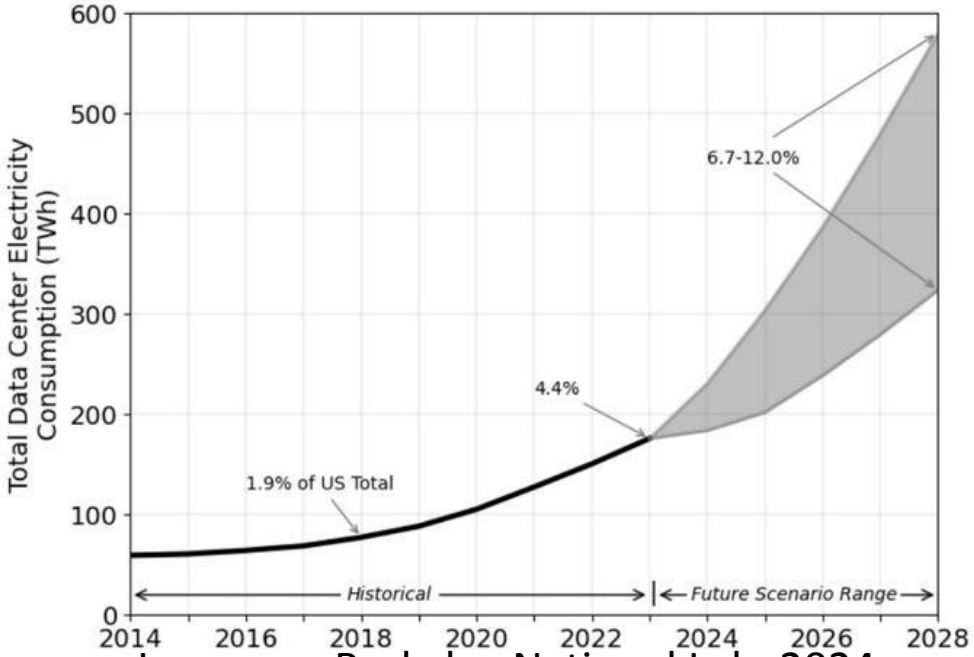
Nature Communications, 2022

AI is having real-world impact

- Public imagination
- Economy Politics
- Law Labor Science
-
-
-
  - Resource use



The New York Times, 2025



Lawrence Berkeley National Lab, 2024

AI is having real-world impact

- Public imagination
- Economy Politics
- Law Labor Science
- Health
-
-
-

The Boston Globe, 2018

AI is having real-world impact

- Public imagination
- Economy Politics
- Law Labor Science
- Health Education

BREAKING

ChatGPT In Schools: Here's Where It's Banned—And How It Could Potentially Help Students

Arianna Johnson Forbes Staff

I cover the latest trends in science, tech and healthcare.

Follow

2

Jan 18, 2023, 02:31pm EST

Forbes, 2023

AI is having real-world impact

- Public imagination
- Economy Politics
- Law Labor Science
- Health Education
- Social interaction
-
-
-

The Washington Post
Democracy Dies in Darkness

‘It’s almost like we never even spoke’: AI is making everyone on dating apps sound charming

As more people use AI to doctor their messages, others are left wondering exactly who they’re falling for.

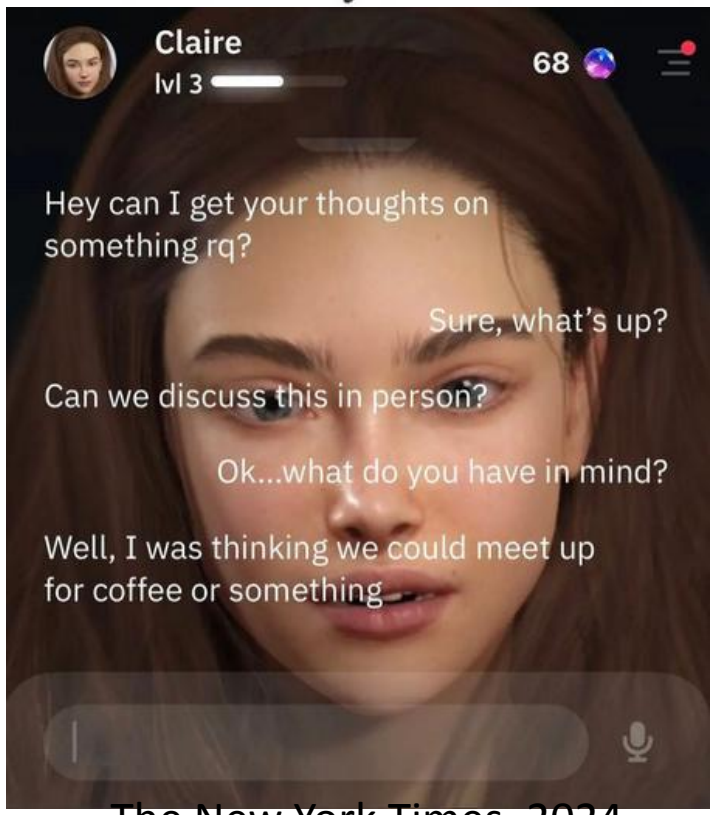
Updated July 3, 2025

AI is having real-world impact

- Public imagination
- Economy Politics
- Law Labor Science
- Health Education
- Social interaction
-
-
-

Meet My A.I. Friends

Our columnist spent the past month hanging out with 18 A.I. companions. They critiqued his clothes, chatted among themselves and hinted at a very different future.



The New York Times, 2024

AI is having real-world impact

- Public imagination
- Economy Politics
- Law Labor Science
- Health Education
- Social interaction
-
-
-

The New York Times, 2025

AI is having real-world impact

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- Social interaction
-
-
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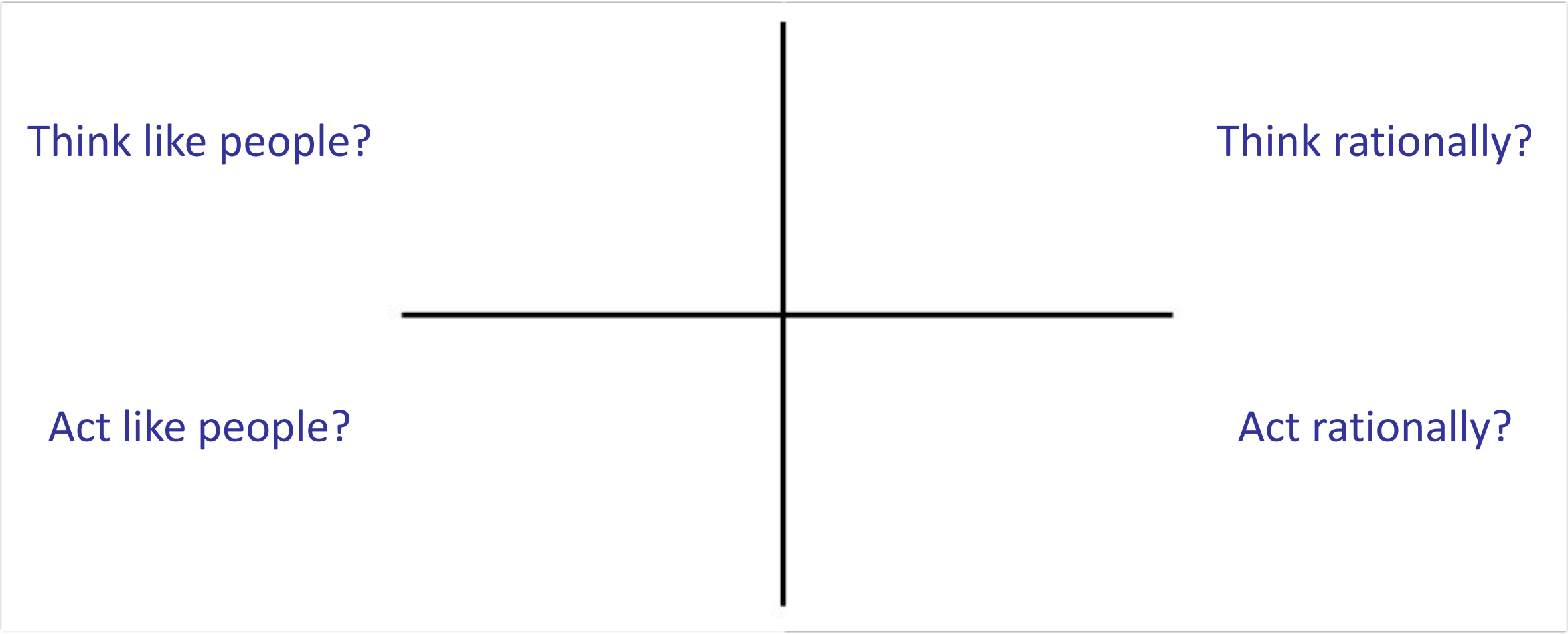
Ok,but what actually is AI???

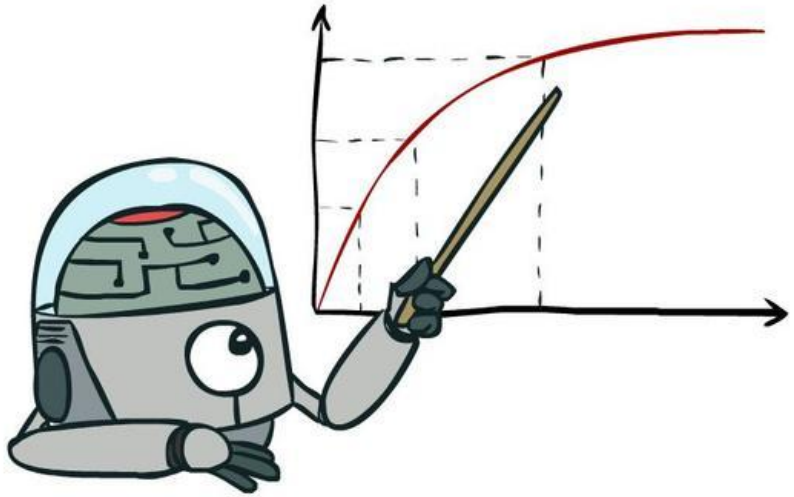
What should we build?

Should we make machines that…



Rational Decisions

- We'll use the term **rational** in a very specific, technical way:
  - Rational: *maximally achieving pre-defined goals* Goals are
  - expressed in terms of the **utility** of outcomes World is
  - uncertain, so we'll use **expected** utility Being rational means
  - acting to **maximize your expected utility**



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A better title for this course might be:

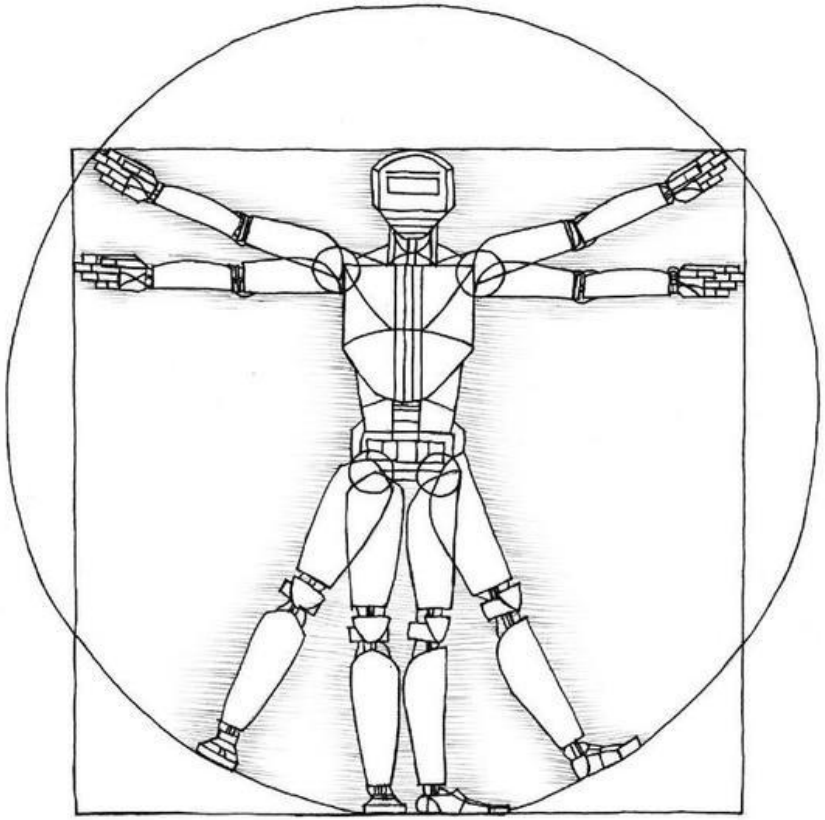
Computational Rationality

Perspectives on Intelligence

- Skills-based perspective “A system is only
- intelligent if it can do [X].”
  - Play chess?
  - Learn from experience?
  - Use words properly?
  - Make mistakes? Not
  - make mistakes?

Perspectives on Intelligence

- Embodiment perspective (Rodney Brooks)



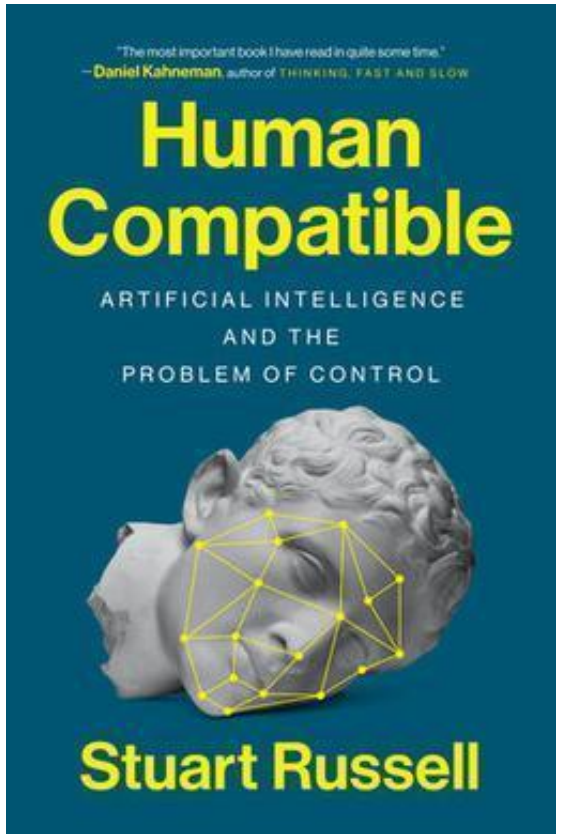
Perspectives on Intelligence

- Psychometrics perspective (François Chollet) “Measuring abilities,
- not skills […] across a broad range of tasks,

including tasks ~~that were previously unknown~~ to the ability-enabled system and its developers.”

Perspectives on Intelligence

- Human-compatible perspective (Stuart Russell)



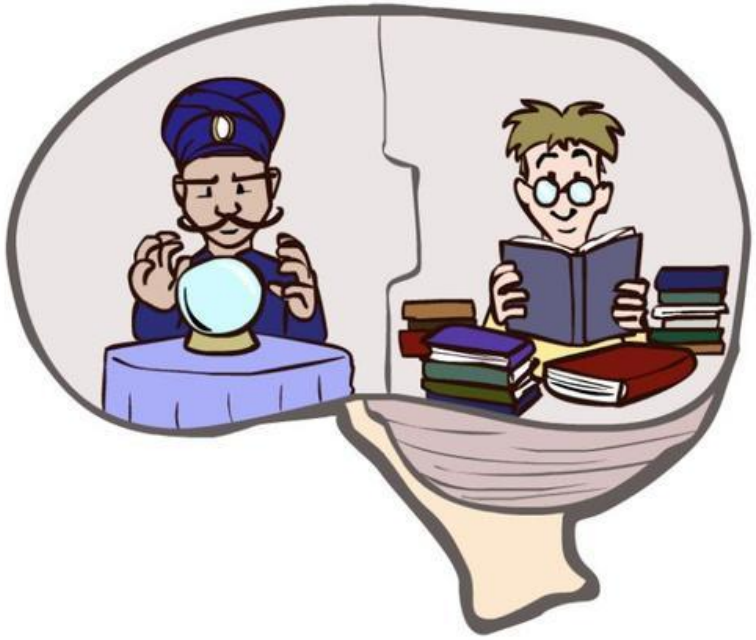
1. Machine's objective is to maximize human utility.
2. Initially uncertain about human preferences.
3. Must learn about preferences from human behavior.

Perspectives on Intelligence

A human being should be able to change a diaper, plan an invasion, butcher a hog, conn a ship, design a building, write a sonnet, balance accounts, build a wall, set a bone, comfort the dying, take orders, give orders, cooperate, act alone, solve equations, analyze a new problem, pitch manure, program a computer, cook a tasty meal, fight efficiently, die gallantly. Specialization is for insects. —Robert A. Heinlein

What About the Brain?

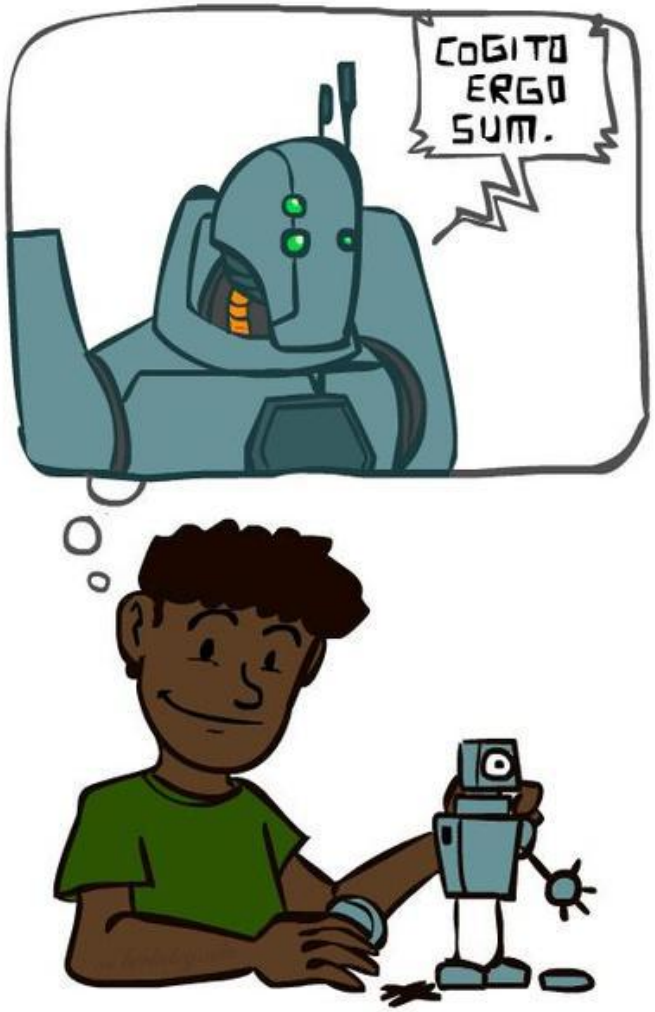
- Brains (human minds) are very good at making rational decisions, but not perfect
- Brains aren't as modular as software, so hard to reverse engineer!
- AI may be better than brains at some tasks



- *"Brains are to intelligence as wings are to flight"*
- We can't yet build AI on the scale of the brain
  - ~100T synapses in the human brain vs ~1.8T weights in GPT4
- Still, the brain can be a great inspiration for AI!

A (Short) History of AI

- 1940-1950: Early days: neural and computer science meet
  - 1943: McCulloch&Pitts:Perceptron–booleancircuitmodelofbrain
  - 1950: Turing's “Computing Machinery and Intelligence”
- 1950—70: Excitement! Logic-driven
  - 1950s: EarlyAIprograms, includingSamuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
  - 1956: Dartmouth meeting: “Artificial Intelligence” adopted



*“We propose that a **2-month, 10-man study of artificial intelligence** be carried out **during the summer of 1956** at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that **every ~~aspect of learning~~ or ~~any other feature of intelligence~~** can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. **We think that a significant advance can be made** in one or more of these problems if a carefully selected group of scientists work on it together for a summer.”*

A (Short) History of AI

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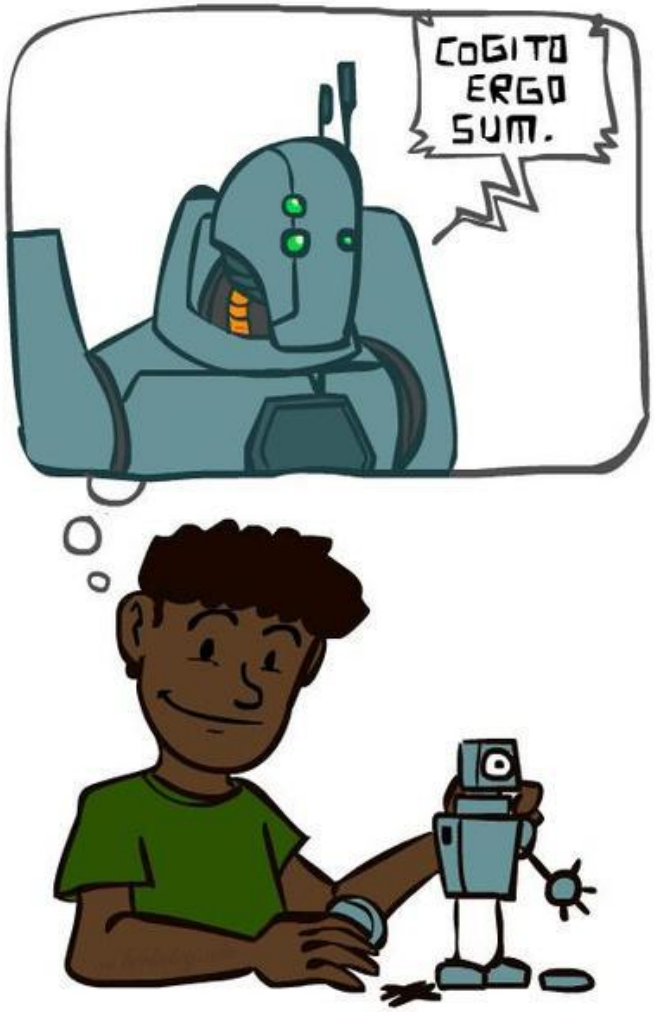
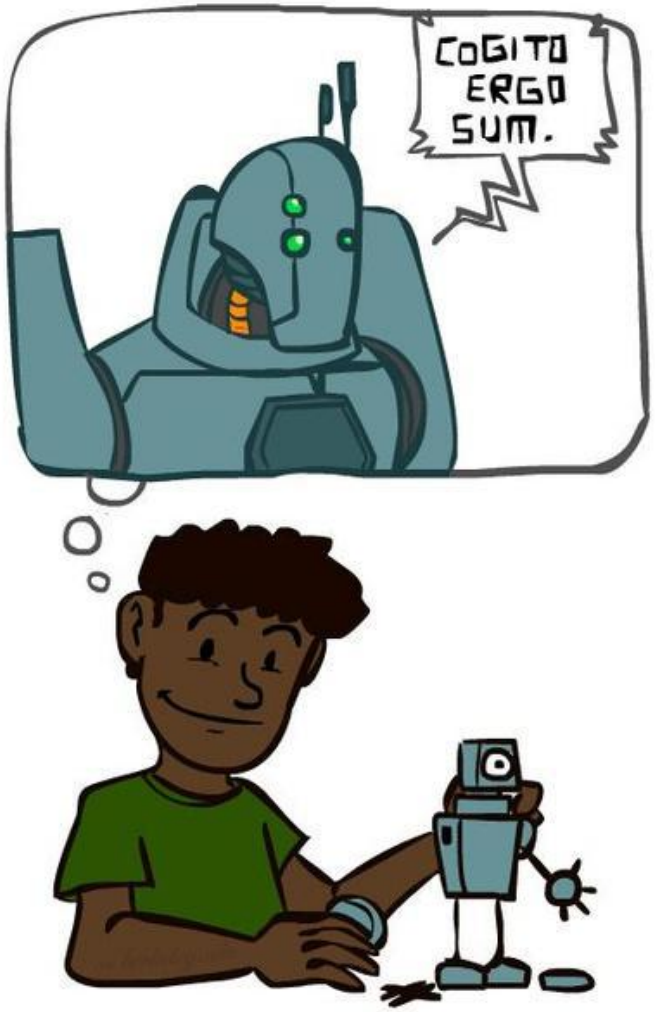
  - 1950s: EarlyAIprograms, includingSamuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
  - 1956: Dartmouth meeting: “Artificial Intelligence” adopted
  - 1969: Minsky & Papert: perceptrons can’t learn XOR/parity!
- 1970—90: Knowledge-based approaches
  - 1969—79: Early developmentofknowledge-based systems
  - 1980—88: Expert systems industry booms; backpropagation makes it feasible to train multi-layer neural networks
  - 1988—93: Expert systems industry busts: “AI Winter”
- 1990—2010: Statistical approaches, agents
  - Resurgence ofprobability,focus on uncertainty
  - Agents and learning systems… “AI Spring”? 1992: TD-Gammon achieves human-level play at backgammon 1997: Deep Blue defeats Gary Kasparov at chess 2002: Embodied AI; Roomba vacuum invented



A (Short) History of AI

- 2010—2017: Big Data, GPUs, Deep Learning
  - 2011: ApplereleasesSiri
  - 2012: AlexNet wins ImageNet competition 2015: DeepMind
  - achieves “human-level” control in Atari games 2016:
  - DeepMind’s AlphaGo defeats Lee Sedol at Go 2016: Google
  - Translate migrates to neural networks
- 2017—: Scaling Up, Large Language Models
  - 2017: Google invents Transformer architecture 2017:
  - DeepStack/Libratus defeat humans at poker 2018-2020: AlphaFold
  - predicts protein structure from amino acids 2021-2022: Modern
  - text-to-image generation 2022: OpenAI releases ChatGPT 2023:
  - Every other company also releases a chatbot 2024: Nobel prizes in
  - physics and chemistry go to AI advances
  -



A (Short) History of AI

- Zooming out, some general patterns:
  - Overclaiming followedby disillusionment (AI winters)
  - Simultaneous discovery, or rediscovery, of ideas
  - Interdisciplinary
    - Math,statistics, controltheory, neuroscience,psychology,economics, philosophy, computer engineering, linguistics
    - And increasingly: law, medicine, education…

What Can AI Do?

Quiz: Which of the following can be done at present? Win against any human at chess? Win against the best humans at Go? Play a decent game of table tennis? Unload any dishwasher in any home? Drive safely along the highway? Drive safely along streets of San Francisco? Buy a week's worth of groceries on the web? Buy a week's worth of groceries at Berkeley Bowl? Discover and prove a new mathematical theorem? Perform a surgical operation? Translate spoken Chinese into spoken English in real time? Win an art competition? Write an intentionally funny story? Construct a building?

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Google's AlphaGo beats Lee Sedol at Go in 2016

But … a plot twist in 2023!

DEEP BLUE WAS JUST THE START —

Man beats machine at *Go* in human victory over AI

Amateur exploited weakness in systems that have otherwise dominated grandmasters.

RICHARD WATERS, FINANCIAL TIMES - 2/19/2023, 4:51 AM

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California just opened the floodgates for self-driving cars

After a state regulatory board ruling, San Francisco will have 24/7 robotaxis on its streets. A Washington Post analysis shows how it's a pivotal moment for the industry.

By Trisha Thadani and Jeremy B. Merrill

Updated August 10, 2023 at 10:06 p.m. EDT | Published August 10, 2023 at 9:15 p.m. EDT

The Register

California DMV hits brakes on Cruise's SF driverless fleet after series of fender benders

50% chop effective 'immediately' as department investigates traffic and safety issues

Jude Karabus

Mon 21 Aug 2023 // 14:28 UTC

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please write me a very funny, and extremely short story

A penguin walks into a bar, looks around, and says,
"Has anyone seen my brother?"

The bartender squints and asks,
"What does he look like?"

ChatGPT, 2025

What Can AI Do?

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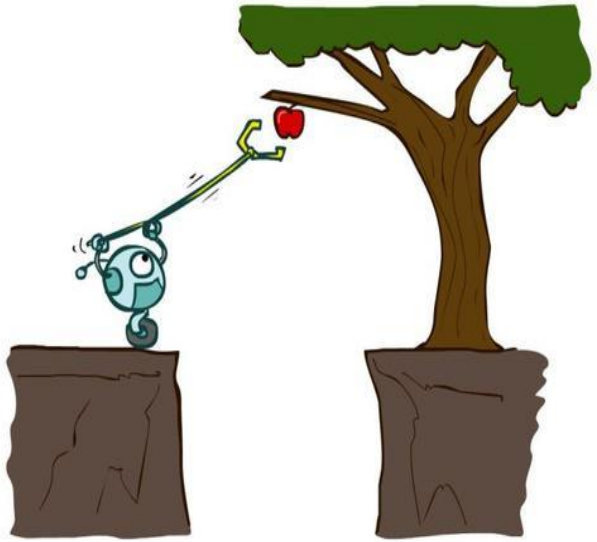
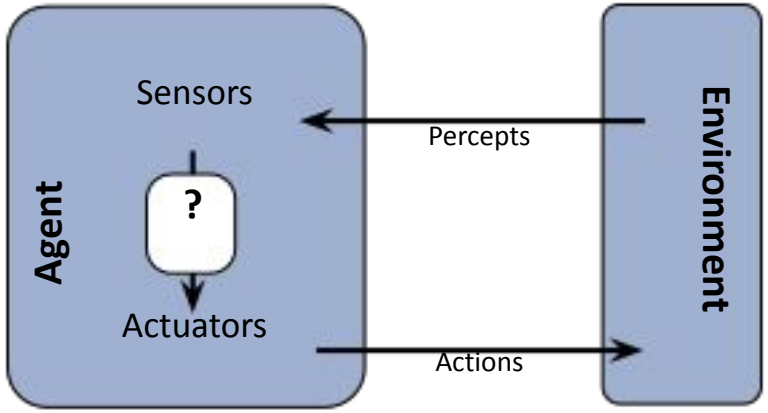
This Course: Designing Rational Agents

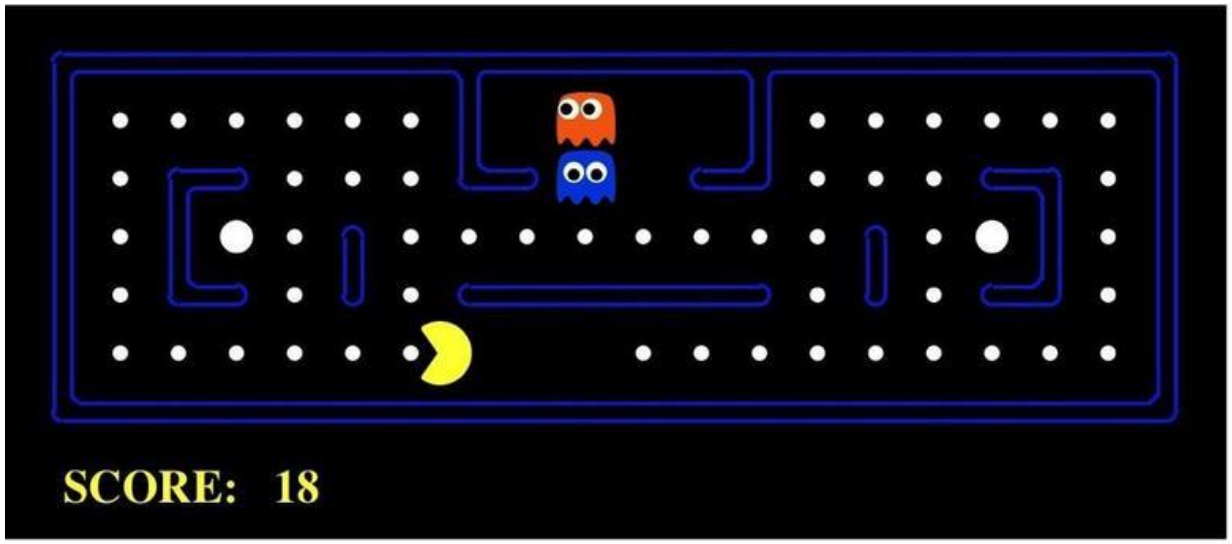
- An **agent** is an entity that perceives and acts. A
- **rational agent** selects actions that maximize
- its (expected) **utility**.
  Characteristics of the **percepts**, **environment**,

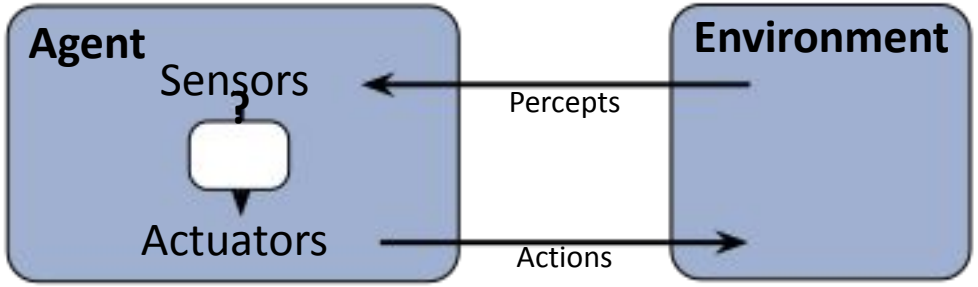
  and **action space** dictate techniques for selecting rational actions
- This course is about:
  - General AI techniques fora varietyof problem types
  - Learning to recognize when and how a new problem can be solved with an existing technique





Pac-Man as an Agent





Pac-Man is a registered trademark of Namco-Bandai Games, used here for educational purposes

Course Topics

Core Components of Rational Agents:

- Search & Planning
- Probability & Inference
- Supervised Learning
- Reinforcement Learning

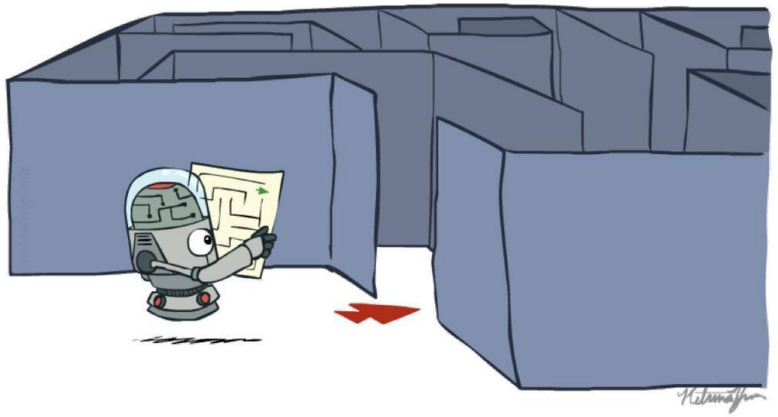
Course Topics

Search & Planning

Probability & Inference

Supervised Learning

Reinforcement Learning



How can I use my ***model*** of the world to find a ***sequence of actions*** to achieve my ***goal***?

Course Topics

Search & Planning

Probability & Inference

Supervised Learning

Reinforcement Learning

How can I make sense of ***uncertainty***?

Course Topics

Search & Planning

Probability & Inference

Supervised Learning

Reinforcement Learning

How can I learn a ***model*** of the world from ***data***?

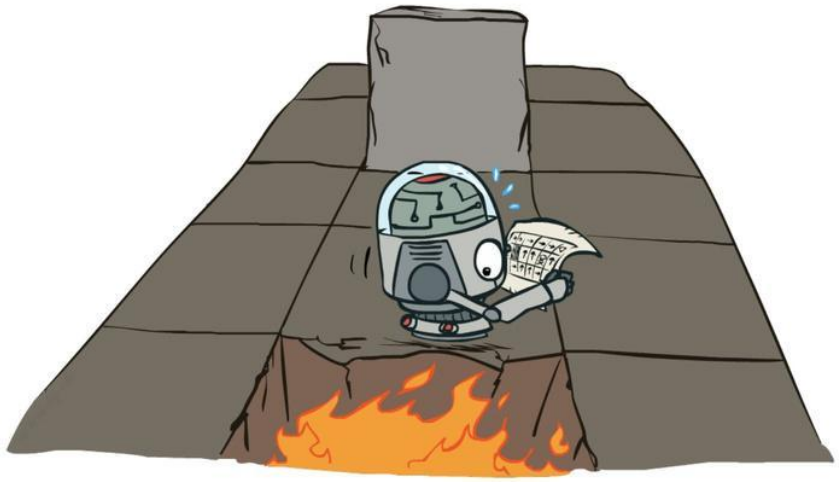
Course Topics

Search & Planning

Probability & Inference

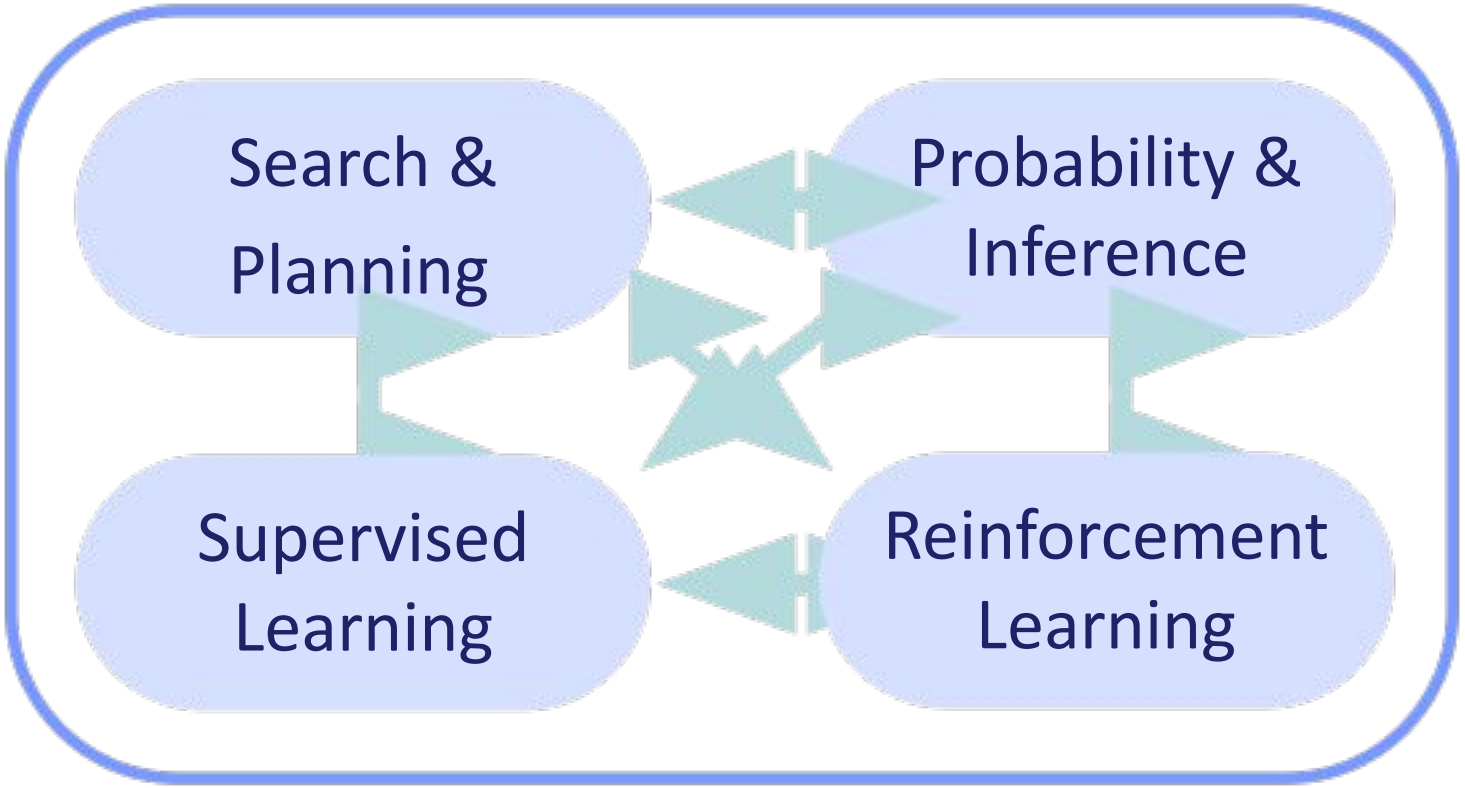
Supervised Learning

Reinforcement Learning

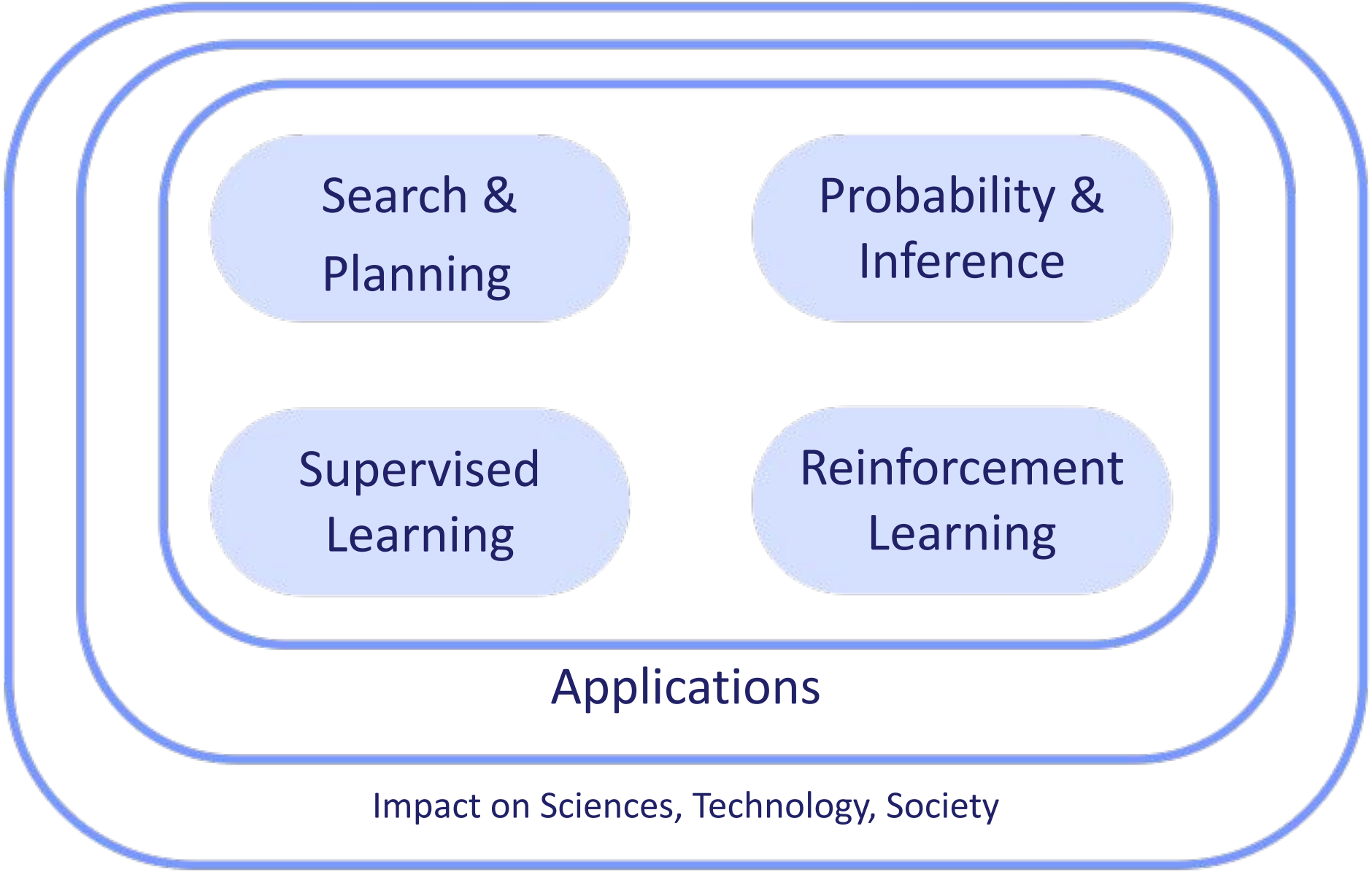


How can I learn a ***policy*** for any situation so that I can ***maximize utility***?

Course Topics



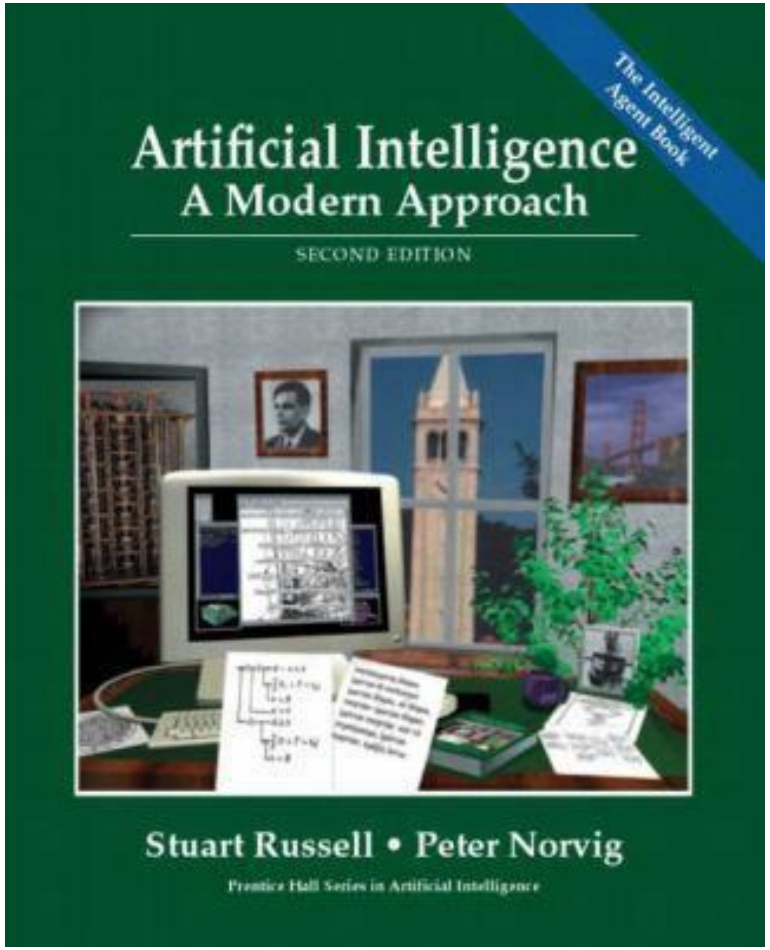
Course Topics



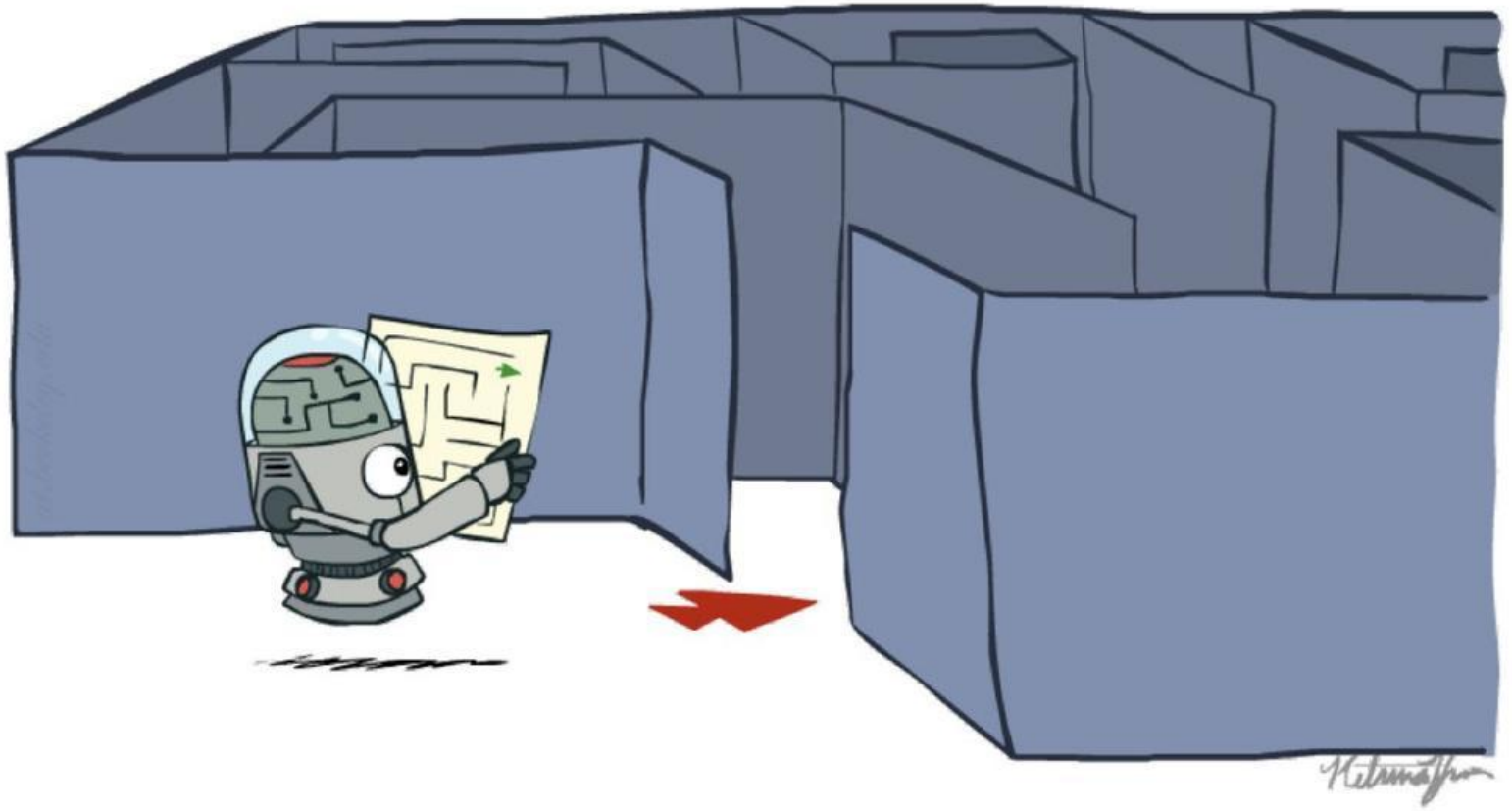
By the end of this course you'll:

- Build and understand math of rational, learning agents Select and
- apply the right AI methods for wide range of problems Recognize
- how these methods are used in modern AI systems Be prepared
- to make decisions on how AI is used in society

The required book for this course



Next Lecture: Search



Quick hits & bits from this lecture…

People to explore

- Yann LeCun – Deep learning researcher, Meta AI leader
- Geoffrey Hinton – Neural networks pioneer
- Yoshua Bengio – AI safety and deep learning researcher → Zero Law nonprofit: https://zerolaw.ai
- Fei-Fei Li – Computer vision leader → World Labs: https://www.worldlabs.ai
- Noam Chomsky – Linguist and cognitive scientist known for work on generative grammar and language theory
- Alan Turing (watch "imitation game" movie, read his iconic paper)

Keywords and lingo

- Hallucination
- Sycophantic
- Loss of consensus reality
- Allignment
- Prompt Engineering
- Prompt Injection
- Agentic AI
- LMM – Language Model